

Vector and Calculus

Single Correct Type Questions

1. When vector $\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ is subtracted from vector $\vec{B}$, it gives a vector equal to $2\hat{j}$. Then the magnitude of vector $\vec{B}$ will be:

[11 April, 2023 (Shift-II)]

- (a) $\sqrt{13}$ (b) $\sqrt{33}$
(c) $\sqrt{6}$ (d) $\sqrt{5}$

2. Match List-I with List-II.

List-I		List-II	
A.	$\vec{C} - \vec{A} - \vec{B} = 0$	I.	
B.	$\vec{A} - \vec{C} - \vec{B} = 0$	II.	
C.	$\vec{B} - \vec{A} - \vec{C} = 0$	III.	
D.	$\vec{A} + \vec{B} = -\vec{C}$	IV.	

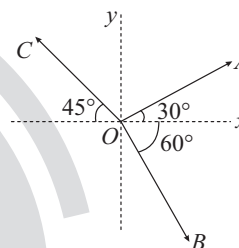
Choose the correct answer from the options given below:

[25 July, 2021 (Shift-I)]

- (a) A-IV, B-I, C-III, D-II
(b) A-IV, B-III, C-I, D-II
(c) A-I, B-IV, C-II, D-III
(d) A-III, B-II, C-IV, D-I

3. The magnitude of vectors $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$ in the given figure are equal. The direction of $\vec{OA} + \vec{OB} - \vec{OC}$ with x-axis will be

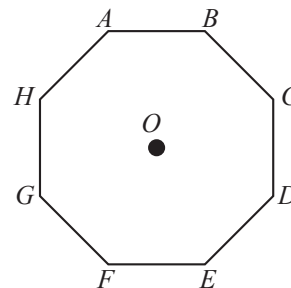
[26 Aug, 2021 (Shift-I)]



- (a) $\tan^{-1} \frac{(\sqrt{3}-1+\sqrt{2})}{(1-\sqrt{3}+\sqrt{2})}$ (b) $\tan^{-1} \frac{(1+\sqrt{3}-\sqrt{2})}{(1-\sqrt{3}-\sqrt{2})}$
(c) $\tan^{-1} \frac{(1-\sqrt{3}-\sqrt{2})}{(1+\sqrt{3}+\sqrt{2})}$ (d) $\tan^{-1} \frac{(\sqrt{3}-1+\sqrt{2})}{(1+\sqrt{3}-\sqrt{2})}$

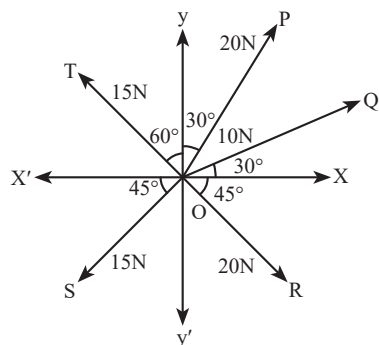
4. In an octagon $ABCDEFGH$ of equal side, what is the sum of $\vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF} + \vec{AG} + \vec{AH}$, if, $\vec{AO} = 2\hat{i} + 3\hat{j} - 4\hat{k}$

[25 Feb, 2021 (Shift-I)]



- (a) $16\hat{i} + 24\hat{j} - 32\hat{k}$
(b) $16\hat{i} - 24\hat{j} + 32\hat{k}$
(c) $-16\hat{i} - 24\hat{j} + 32\hat{k}$
(d) $16\hat{i} + 24\hat{j} - 32\hat{k}$

5. The resultant of these forces $\overrightarrow{OP}$, $\overrightarrow{OQ}$, $\overrightarrow{OR}$, $\overrightarrow{OS}$ and $\overrightarrow{OT}$ is approximately N. [27 Aug, 2021 (Shift-I)]
[Take $\sqrt{3} = 1.7$, $\sqrt{2} = 1.4$. Given $\hat{i}$ and $\hat{j}$ unit vectors along x, y axis]



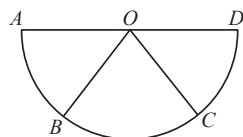
- (a) $9.25\hat{i} + 5\hat{j}$ (b) $2.5\hat{i} - 14.5\hat{j}$
(c) $-1.5\hat{i} - 15.5\hat{j}$ (d) $3\hat{i} + 15\hat{j}$
6. Two forces having magnitude A and $\frac{A}{2}$ are perpendicular to each other. The magnitude of their resultant is [8 April, 2023 (Shift-I)]
- (a) $\frac{\sqrt{5}A}{4}$ (b) $\frac{5A}{2}$
(c) $\frac{\sqrt{5}A^2}{2}$ (d) $\frac{\sqrt{5}A}{2}$
7. A vector in x - y plane makes an angle of 30° with y -axis. The magnitude of y -component of vector is $2\sqrt{3}$. The magnitude of x -component of the vector will be: [15 April, 2023 (Shift-I)]

- (a) $\frac{1}{\sqrt{3}}$ (b) 6
(c) $\sqrt{3}$ (d) 2
8. **Assertion A:** If A , B , C and D are four points on a semicircular arc with centre at ' O ' such that $|\overrightarrow{AB}| = |\overrightarrow{BC}| = |\overrightarrow{CD}|$, then

$$\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} = 4\overrightarrow{AO} + \overrightarrow{OB} + \overrightarrow{OC}$$

Reason R: Polygon law of vector addition yields

$$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} = \overrightarrow{AD} = 2\overrightarrow{AO}$$

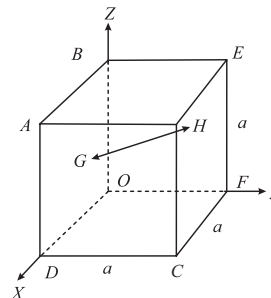


In the light of the above statements, choose the most appropriate answer from the options given below:

[27 July, 2021 (Shift-I)]

- (a) **A** is not correct but **R** is correct.
(b) **A** is correct but **R** is not correct.
(c) Both **A** and **R** are correct and **R** is the correct explanation of **A**.
(d) Both **A** and **R** are correct but **R** is not the correct explanation of **A**.

9. In the cube of side ' a ' shown in the figure, the vector from the central point of the face $ABOD$ to the central point of the face $BEFO$ will be [10 Jan, 2019 (Shift-I)]



- (a) $\frac{1}{2}a(\hat{k} - \hat{i})$ (b) $\frac{1}{2}a(\hat{i} - \hat{k})$
(c) $\frac{1}{2}a(\hat{j} - \hat{i})$ (d) $\frac{1}{2}a(\hat{j} - \hat{k})$
10. If two vectors $\vec{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\vec{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other. Then, the value of m will be: [24 Jan, 2023 (Shift-II)]
- (a) 1 (b) -1
(c) -3 (d) 2

11. Which of the following relations is true for two unit vector $\hat{A}$ and $\hat{B}$ making an angle to θ each other? [25 June, 2022 (Shift-I)]

- (a) $|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \tan \frac{\theta}{2}$
(b) $|\hat{A} - \hat{B}| = |\hat{A} + \hat{B}| \tan \frac{\theta}{2}$
(c) $|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \cos \frac{\theta}{2}$
(d) $|\hat{A} - \hat{B}| = |\hat{A} + \hat{B}| \cos \frac{\theta}{2}$

12. Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. If magnitude of $\vec{A} + \vec{B}$ is equal to two times the magnitude of $\vec{A} - \vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ will be:

[29 June, 2022 (Shift-I)]

- (a) $\sin^{-1}\left(\frac{3}{5}\right)$ (b) $\sin^{-1}\left(\frac{1}{3}\right)$
(c) $\cos^{-1}\left(\frac{3}{5}\right)$ (d) $\cos^{-1}\left(\frac{1}{3}\right)$

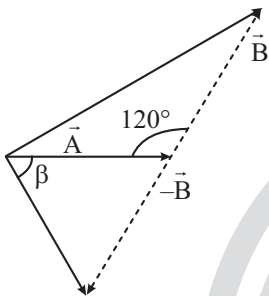
13. If $\vec{A}$ and $\vec{B}$ are two vectors satisfying the relation $\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}|$. Then the value of $|\vec{A} - \vec{B}|$ will be:

[20 July, 2021 (Shift-I)]

- (a) $\sqrt{A^2 + B^2 - \sqrt{2}AB}$
 (b) $\sqrt{A^2 + B^2}$
 (c) $\sqrt{A^2 + B^2 + 2AB}$
 (d) $\sqrt{A^2 + B^2 + \sqrt{2}AB}$

14. The angle between vector $(\vec{A})$ and $(\vec{A} - \vec{B})$ is:

[26 Aug, 2021 (Shift-II)]



- (a) $\tan^{-1}\left(\frac{\sqrt{3}B}{2A - B}\right)$
 (b) $\tan^{-1}\left(\frac{B \cos \theta}{A - B \sin \theta}\right)$
 (c) $\tan^{-1}\left(\frac{A}{0.7 B}\right)$
 (d) $\tan^{-1}\left(\frac{-\frac{B}{2}}{A - B \frac{\sqrt{3}}{2}}\right)$

15. What will be the projection of vector $\vec{A} = \hat{i} + \hat{j} + \hat{k}$ on vector $\vec{B} = \hat{i} + \hat{j}$?

[22 July, 2021 (Shift-II)]

- (a) $2(\hat{i} + \hat{j} + \hat{k})$ (b) $\sqrt{2}(\hat{i} + \hat{j})$
 (c) $(\hat{i} + \hat{j})$ (d) $\sqrt{2}(\hat{i} + \hat{j} + \hat{k})$

16. Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. The magnitude of $(\vec{A} + \vec{B})$ is 'n' times the magnitude of $(\vec{A} - \vec{B})$. The angle between $\vec{A}$ and $\vec{B}$ is:

- (a) $\cos^{-1}\left[\frac{n^2 - 1}{n^2 + 1}\right]$ (b) $\cos^{-1}\left[\frac{n - 1}{n + 1}\right]$
 (c) $\sin^{-1}\left[\frac{n^2 - 1}{n^2 + 1}\right]$ (d) $\sin^{-1}\left[\frac{n - 1}{n + 1}\right]$

17. Two vectors $\vec{X}$ and $\vec{Y}$ have equal magnitude. The magnitude of $(\vec{X} - \vec{Y})$ is n times the magnitude of $(\vec{X} + \vec{Y})$. The angle between $\vec{X}$ and $\vec{Y}$ is:

[25 July, 2021 (Shift-II)]

- (a) $\cos^{-1}\left(\frac{n^2 + 1}{n^2 - 1}\right)$ (b) $\cos^{-1}\left(\frac{n^2 - 1}{n^2 + 1}\right)$
 (c) $\cos^{-1}\left(\frac{n^2 - 1}{-n^2 - 1}\right)$ (d) $\cos^{-1}\left(\frac{n^2 + 1}{-n^2 - 1}\right)$

18. Two vectors $\vec{P}$ and $\vec{Q}$ have equal magnitudes. If the magnitude of $\vec{P} + \vec{Q}$ is n times the magnitude of $\vec{P} - \vec{Q}$, then angle between $\vec{P}$ and $\vec{Q}$ is:

- (a) $\sin^{-1}\left(\frac{n^2 - 1}{n^2 + 1}\right)$ (b) $\cos^{-1}\left(\frac{n^2 - 1}{n^2 + 1}\right)$
 (c) $\sin^{-1}\left(\frac{n - 1}{n + 1}\right)$ (d) $\cos^{-1}\left(\frac{n - 1}{n + 1}\right)$

19. **Statement-I:** Two forces $(\vec{P} + \vec{Q})$ and $(\vec{P} - \vec{Q})$ where $\vec{P} \perp \vec{Q}$, when act at an angle θ_1 to each other, the magnitude of their resultant is $\sqrt{3(P^2 + Q^2)}$, when they act at an angle θ_2 , the magnitude of their resultant becomes $\sqrt{2(P^2 + Q^2)}$. This is possible only when $\theta_1 < \theta_2$.

[31 Aug, 2021 (Shift-II)]

Statement-II: In the situation given above. $\theta_1 = 60^\circ$ and $\theta_2 = 90^\circ$

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both Statement-I and Statement-II are true
 (b) Both Statement-I and Statement-II are false
 (c) Statement-I is true but Statement-II is false
 (d) Statement-I is false but Statement-II is true

20. Let $|\vec{A}_1| = 3$, $|\vec{A}_2| = 5$ and $|\vec{A}_1 + \vec{A}_2| = 5$. The value of $(2\vec{A}_1 + 3\vec{A}_2) \cdot (3\vec{A}_1 - 2\vec{A}_2)$ is:

[8 April, 2019 (Shift-II)]

- (a) -112.5 (b) -106.5
 (c) -118.5 (d) -99.5

21. Two forces P and Q, of magnitude 2F and 3F, respectively, are at an angle θ with each other. If the force Q is doubled, then their resultant also gets doubled. Then, the angle θ is:

[10 Jan, 2019 (Shift-II)]

(a) 120° (b) 60°

(c) 90° (d) 30°

22. $\vec{A}$ is a vector quantity such that $|\vec{A}| = \text{non-zero constant}$.

Which of the following expression is true for $\vec{A}$?

[25 June, 2022 (Shift-I)]

(a) $\vec{A} \cdot \vec{A} = 0$

(b) $\vec{A} \times \vec{A} < 0$

(c) $\vec{A} \times \vec{A} = 0$

(d) $\vec{A} \times \vec{A} > 0$

23. If force $\vec{F} = 3\hat{i} + 4\hat{j} - 2\hat{k}$ acts on a particle having position vector $2\hat{i} + \hat{j} + 2\hat{k}$ then, the torque about the origin will be:

[25 June, 2022 (Shift-I)]

(a) $3\hat{i} + 4\hat{j} - 2\hat{k}$

(b) $-10\hat{i} + 10\hat{j} + 5\hat{k}$

(c) $10\hat{i} + 5\hat{j} - 10\hat{k}$

(d) $10\hat{i} + \hat{j} - 5\hat{k}$

24. If $t = \sqrt{x} + 4$, then $\left(\frac{dx}{dt}\right)_{t=4}$ is: [29 July, 2022 (Shift-I)]

(a) 4

(b) zero

(c) 8

(d) 16

Integer Type Questions

25. Vectors $a\hat{i} + b\hat{j} + \hat{k}$ and $2\hat{i} - 3\hat{j} + 4\hat{k}$ are perpendicular to each other when $3a + 2b = 7$, the ratio of a to b is $\frac{x}{2}$. The value of x is _____. [24 Jan, 2023 (Shift-I)]

26. If $\vec{A} = (2\hat{i} + 3\hat{j} - \hat{k})m$ and $\vec{B} = (\hat{i} + 2\hat{j} - 2\hat{k})m$. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be _____ m. [26 July, 2022 (Shift-II)]

27. If the projection of $2\hat{i} + 4\hat{j} - 2\hat{k}$ on $\hat{i} + 2\hat{j} + \alpha\hat{k}$ is zero. Then, the value of α will be _____. [28 July, 2022 (Shift-I)]

28. If $\vec{P} = 3\hat{i} + \sqrt{3}\hat{j} + 2\hat{k}$ and $\vec{Q} = 4\hat{i} + \sqrt{3}\hat{j} + 2.5\hat{k}$ then, the unit vector in the direction of $\vec{P} \times \vec{Q}$ is $\frac{1}{x}(\sqrt{3}\hat{i} + \hat{j} - 2\sqrt{3}\hat{k})$. The value of x is _____. [25 Jan, 2023 (Shift-I)]

29. If $\vec{P} \times \vec{Q} = \vec{Q} \times \vec{P}$ the angle between $\vec{P}$ and $\vec{Q}$ is θ ($0^\circ < \theta < 360^\circ$). The value of ' θ ' will be _____. [25 Feb, 2021 (Shift-II)]

30. A person of height 1.6 m is walking away from a lamp post of height 4m along a straight path on the flat ground. The lamp post and the person are always perpendicular to the ground. If the speed of the person is 60 cm s^{-1} , the speed of the tip of the person's shadow on the ground with respect to the person is _____ cm s^{-1} .

[JEE Adv, 2023]

ANSWER KEY

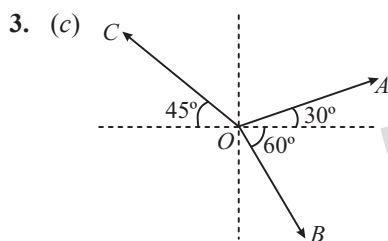
1. (b) 2. (b) 3. (c) 4. (a) 5. (a) 6. (d) 7. (d) 8. (d) 9. (c) 10. (d)
 11. (b) 12. (c) 13. (a) 14. (a) 15. (c) 16. (a) 17. (c) 18. (b) 19. (a) 20. (c)
 21. (a) 22. (c) 23. (b) 24. (b) 25. [1] 26. [2] 27. [5] 28. [4] 29. [180] 30. [40]

EXPLANATIONS

1. (b) Given, $\vec{B} - \vec{A} = 2\hat{j}$, $\vec{B} = 2\hat{j} + 2\hat{i} + 3\hat{j} + 2\hat{k}$
 $\therefore \vec{B} = 2\hat{i} + 5\hat{j} + 2\hat{k}$

$$|\vec{B}| = \sqrt{(2)^2 + (5)^2 + (2)^2} = \sqrt{33} \quad (\text{Bonus})$$

2. (b) Apply the triangle law of addition, to get answer.



According to diagram, let magnitude be equal to

$$\vec{OA} = \lambda \left[\cos 30^\circ \hat{i} + \sin 30^\circ \hat{j} \right] = \lambda \left[\frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j} \right]$$

$$\vec{OB} = \lambda \left[\cos 60^\circ \hat{i} - \sin 60^\circ \hat{j} \right] = \lambda \left[\frac{1}{2} \hat{i} - \frac{\sqrt{3}}{2} \hat{j} \right]$$

$$\vec{OC} = \lambda \left[\cos 45^\circ (-\hat{i}) + \sin 45^\circ \hat{j} \right] = \lambda \left[-\frac{1}{\sqrt{2}} \hat{i} + \frac{1}{\sqrt{2}} \hat{j} \right]$$

$$\therefore \vec{OA} + \vec{OB} - \vec{OC}$$

$$= \lambda \left[\left(\frac{\sqrt{3}+1}{2} + \frac{1}{\sqrt{2}} \right) \hat{i} + \left(\frac{1}{2} - \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \right) \hat{j} \right]$$

$$\tan^{-1} \left[\frac{\frac{1}{2} - \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}}{\frac{\sqrt{3}+1}{2} + \frac{1}{\sqrt{2}}} \right] = \tan^{-1} \left[\frac{\sqrt{2} - \sqrt{6} - 2}{\sqrt{6} + \sqrt{2} + 2} \right]$$

$$= \tan^{-1} \left[\frac{1 - \sqrt{3} - \sqrt{2}}{\sqrt{3} + 1 + \sqrt{2}} \right]$$

4. (a) $\vec{AB} + \vec{AH} = \vec{AO}$

$$\vec{AC} + \vec{AG} = 2\vec{AO}$$

$$\vec{AD} + \vec{AF} = 3\vec{AO}$$

$$\vec{AE} = 2\vec{AO}$$

Adding all,

$$\vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF} + \vec{AG} + \vec{AH} = 8\vec{AO}$$

$$= 16\hat{i} + 24\hat{j} - 32\hat{k}$$

5. (a) The horizontal component of force,

$$\vec{F}_x = \left[10 \times \frac{\sqrt{3}}{2} + 20 \times \frac{1}{2} + \frac{20}{\sqrt{2}} - \frac{15}{\sqrt{2}} - \frac{15\sqrt{3}}{2} \right] = 9.25\hat{i}$$

The vertical component of force,

$$\vec{F}_y = \left[15 \times \frac{1}{2} + 20 \times \frac{\sqrt{3}}{2} + 10 \times \frac{1}{2} - \frac{15}{\sqrt{2}} - \frac{20}{\sqrt{2}} \right] = 5\hat{j}$$

$$\therefore \vec{F}_R = \vec{F}_x + \vec{F}_y = 9.25\hat{i} + 5\hat{j}$$

6. (d) $\vec{F} = (\vec{F}_1 + \vec{F}_2)$

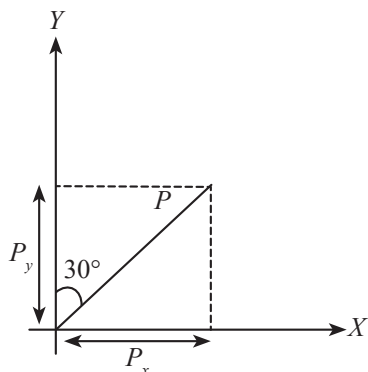
$$|\vec{F}_1| = A, |\vec{F}_2| = \frac{A}{2}$$

$$\theta = 90^\circ$$

$$\therefore |\vec{F}| = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$$

$$|\vec{F}| = \sqrt{A^2 + \frac{A^2}{4}} = \frac{A\sqrt{5}}{2}$$

7. (d) Let the vector be P.



$$P_y = P \cos 30^\circ = 2\sqrt{3}$$

$$\Rightarrow P \frac{\sqrt{3}}{2} = 2\sqrt{3} \Rightarrow P = 4$$

$$\text{Now } P_x = P \sin 30^\circ = 4 \times \frac{1}{2} = 2$$

8. (d) Polygon law is applicable in both the cases given but the equation given in the reason is not useful in explaining the assertion.

$$9. (c) \quad G \equiv \left(\frac{a}{2}, 0, \frac{a}{2} \right), \quad H \equiv \left(0, \frac{a}{2}, \frac{a}{2} \right)$$

$$\vec{GH} = -\frac{a}{2}\hat{i} + \frac{a}{2}\hat{j} = \frac{a}{2}(\hat{j} - \hat{i})$$

10. (d) As $\vec{P}$ and $\vec{Q}$ are perpendicular their dot product must be zero.

$$\vec{P} \cdot \vec{Q} = 0$$

$$(\hat{i} + 2m\hat{j} + m\hat{k}) \cdot (4\hat{i} - 2\hat{j} + m\hat{k}) = 0$$

$$\Rightarrow 4 - 4m + m^2 = 0$$

$$\Rightarrow m = 2$$

$$11. (b) \quad |\vec{A} + \vec{B}| = \sqrt{|\vec{A}|^2 + |\vec{B}|^2 + 2|\vec{A}||\vec{B}|\cos\theta}$$

$$= \sqrt{1+1+2\cos\theta}$$

$$= 2 \cos \theta/2$$

$$|\vec{A} - \vec{B}| = \sqrt{|\vec{A}|^2 + |\vec{B}|^2 - 2|\vec{A}||\vec{B}|\cos\theta}$$

$$= \sqrt{1-2\cos\theta} = 2 \sin \theta/2$$

$$\frac{|\vec{A} + \vec{B}|}{|\vec{A} - \vec{B}|} = \cot \theta/2 \Rightarrow |\vec{A} - \vec{B}| = |\vec{A} + \vec{B}| \tan \frac{\theta}{2}$$

12. (c) According to the question $|\vec{A} + \vec{B}| = 2|\vec{A} - \vec{B}| \dots (i)$

$$\text{It is given that, } |\vec{A}| = |\vec{B}|$$

Squaring equation (i) both side. $[\because A = B]$

$$|\vec{A} + \vec{B}|^2 = (2|\vec{A} - \vec{B}|)^2$$

$$\Rightarrow A^2 + B^2 + 2\vec{A} \cdot \vec{B} = 4(A^2 + B^2 - 2\vec{A} \cdot \vec{B})$$

$$\Rightarrow 2A^2 + 2A^2 \cos\theta = 4(A^2 - B^2 - A^2 \cos\theta)$$

$$\Rightarrow 2A^2 + 2A^2 \cos\theta = 8A^2 - 8A^2 \cos\theta$$

$$\Rightarrow 10A^2 \cos\theta = 6A^2 \Rightarrow \cos\theta = \frac{3}{5}$$

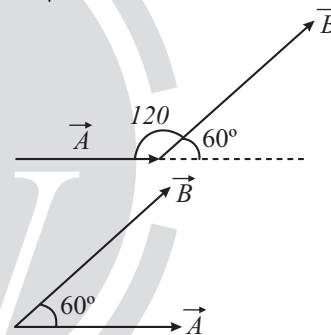
$$\therefore \theta = \cos^{-1}(3/5)$$

13. (a) Given that, $\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}| \Rightarrow AB \cos \theta = AB \sin \theta$
 $\Rightarrow \theta = 45^\circ$

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos \theta}$$

$$= \sqrt{A^2 + B^2 - 2AB \times \frac{1}{\sqrt{2}}} = \sqrt{A^2 + B^2 - \sqrt{2}AB}$$

14. (a)



According to diagram,

Angle between $\vec{A}$ and $\vec{B}$, $\theta = 60^\circ$

Angle between $\vec{A}$ and $-\vec{B}$, $\theta = 120^\circ$

If angle between $\vec{A}$ and $\vec{A} - \vec{B}$ is α

$$\text{then } \tan \alpha = \frac{|-\vec{B}| \sin \theta}{|\vec{A}| + |-\vec{B}| \cos \theta}$$

$$= \frac{B \sin 120^\circ}{A + B \cos 120^\circ}$$

$$= \frac{B \frac{\sqrt{3}}{2}}{A - \frac{B}{2}}$$

$$\Rightarrow \tan \alpha = \frac{\sqrt{3}B}{2A - B}$$

Hence, the angle between vector $\vec{A}$ and $(\vec{A} - \vec{B})$ is

$$\tan \alpha = \frac{\sqrt{3}B}{2A-B}$$

15. (c) Component of $\vec{A}$ an $\vec{B}$

$$\text{Projection of } \vec{A} \text{ on } \vec{B} = (\vec{A} \cdot \hat{B})\hat{B}$$

$$= \left[(\hat{i} + \hat{j} + \hat{k}) \cdot \frac{(\hat{i} + \hat{j})}{\sqrt{2}} \right] \frac{(\hat{i} + \hat{j})}{\sqrt{2}}$$

$$= \frac{1}{2}(1+1)(\hat{i} + \hat{j}) = \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) \frac{(\hat{i} + \hat{j})}{\sqrt{2}} = \hat{i} + \hat{j}$$

16. (a) $|\vec{A} + \vec{B}| = n |\vec{A} - \vec{B}|$

$$\Rightarrow A^2 + B^2 + 2AB \cos \theta$$

$$= n^2(A^2 + B^2 - 2AB \cos \theta)$$

$$\Rightarrow \cos \theta(1 + n^2) = \frac{2a^2(n^2 - 1)}{2a^2} [A = B = a]$$

$$\cos \theta = \frac{n^2 - 1}{n^2 + 1}$$

17. (c) Given,

$$|\vec{x}| = |\vec{y}| \text{ and } |\vec{x} - \vec{y}| = n |\vec{x} + \vec{y}|$$

$$x^2 + y^2 - 2\vec{x} \cdot \vec{y} = n^2(x^2 + y^2 + 2\vec{x} \cdot \vec{y})$$

$$(1 - n^2)(x^2 + y^2) = (1 + n^2)2\vec{x} \cdot \vec{y}$$

$$(1 - n^2)(x^2 + y^2) = (1 + n^2)2xy \cos \theta$$

$$\cos \theta = \frac{1 - n^2}{1 + n^2}$$

$$\theta = \cos^{-1} \left(\frac{n^2 - 1}{-n^2 - 1} \right)$$

18. (b) It is given that, $|\vec{P}| = |\vec{Q}|$

$$\text{Let us suppose, } |\vec{P}| = |\vec{Q}| = x$$

Let the angle between $\vec{P}$ and $\vec{Q}$ is θ , so according to question, we can write

$$|\vec{P} + \vec{Q}| = n |\vec{P} - \vec{Q}| \Rightarrow \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

$$= n \sqrt{P^2 + Q^2 - 2PQ \cos \theta}$$

$$\Rightarrow \sqrt{x^2 + x^2 + 2x^2 \cos \theta} = n \sqrt{x^2 + x^2 - 2x^2 \cos \theta}$$

$$\Rightarrow 2 + 2 \cos \theta = 2n^2 - 2n^2 \cos \theta \Rightarrow \cos \theta = \frac{n^2 - 1}{n^2 + 1}$$

$$\Rightarrow \theta = \cos^{-1} \frac{n^2 - 1}{n^2 + 1}$$

19. (a) According to question the given data is two forces

$$(\vec{P} + \vec{Q}) \text{ and } (\vec{P} - \vec{Q})$$

$$\vec{A} = \vec{P} + \vec{Q} \Rightarrow \vec{B} = \vec{P} - \vec{Q} \Rightarrow \vec{P} \perp \vec{Q}$$

$$\Rightarrow |\vec{A}| = |\vec{B}| = \sqrt{2(P^2 + Q^2)}(1 + \cos \theta)$$

$$\Rightarrow \text{For } |\vec{A} + \vec{B}| = \sqrt{3(P^2 + Q^2)} \Rightarrow \theta_1 = 60^\circ$$

$$\Rightarrow \text{For } |\vec{A} - \vec{B}| = \sqrt{2(P^2 + Q^2)} \Rightarrow \theta_2 = 90^\circ$$

According to solution Both Statement-I and Statement-II are true.

20. (c) $|\vec{A}_1| = 3, |\vec{A}_2| = 5, |\vec{A}_1 + \vec{A}_2| = 5$

$$|\vec{A}_1 + \vec{A}_2| = \sqrt{|\vec{A}_1|^2 + |\vec{A}_2|^2 + 2|\vec{A}_1||\vec{A}_2|\cos \theta}$$

$$5 = \sqrt{9 + 25 + 2 \times 3 \times 5 \cos \theta}$$

$$\Rightarrow \cos \theta = -\frac{9}{2 \times 3 \times 5} = -\frac{3}{10}$$

$$(2\vec{A}_1 + 3\vec{A}_2) \cdot (3\vec{A}_1 - 2\vec{A}_2)$$

$$= 6|\vec{A}_1|^2 + 9\vec{A}_1 \cdot \vec{A}_2 - 4\vec{A}_1 \cdot \vec{A}_2 - 6|\vec{A}_2|^2 = -118.5$$

21. (a) $2|\vec{P} + \vec{Q}| = |\vec{P} + 2\vec{Q}|$

$$\Rightarrow 13 + 12 \cos \theta = 10 + 6 \cos \theta$$

$$\cos \theta = -\frac{1}{2}$$

$$\theta = 120^\circ$$

22. (c) Given, $|A| \neq 0$

$$\vec{A} \times \vec{A} = |A| |A| \sin \theta \hat{n}$$

$$= |A| |A| \sin 0^\circ \hat{n}$$

$$= 0 [\text{Since Angle between the vectors are zero degree}]$$

$$\vec{A} \times \vec{A} = 0$$

23. (b) $\vec{\tau} = \vec{r} \times \vec{F}$

$$= (2\hat{i} + \hat{j} + 2\hat{k}) \times (3\hat{i} + 4\hat{j} - 2\hat{k}) = \begin{vmatrix} \hat{i} & -\hat{j} & \hat{k} \\ 2 & 1 & 2 \\ 3 & 4 & -2 \end{vmatrix}$$

$$= \hat{i}(-2-8) - \hat{j}(-4-6) + \hat{k}(8-3)$$

$$\vec{\tau} = -10\hat{i} + 10\hat{j} + 5\hat{k}$$

24. (b) $t = \sqrt{x} + 4$, at $t = 4, \sqrt{x} = 0$

So, $\frac{dt}{dx} = \frac{1}{2\sqrt{x}} \therefore \left(\frac{dx}{dt}\right)_{t=4} = (2\sqrt{x})_{t=4} = 0$

25. [1] For two perpendicular vectors

$$(a\hat{i} + b\hat{j} + \hat{k}) \cdot (2\hat{i} - 3\hat{j} + 4\hat{k}) = 0$$

$$2a - 3b + 4 = 0$$

On solving, $2a - 3b = -4$

Also given

$$3a + 2b = 7$$

We get $a = 1, b = 2$

$$\frac{a}{b} = \frac{x}{2} \Rightarrow x = \frac{2a}{b} = \frac{2 \times 1}{2}$$

$$\Rightarrow x = 1$$

26. [2] The magnitude of the component of vector A along vector B will be

$$= \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} = \frac{2 + 6 - 2}{3} = 2$$

27. [5] $0 = (2\hat{i} + 4\hat{j} - 2\hat{k}) \cdot \left(\frac{\hat{i} + 2\hat{j} + \alpha\hat{k}}{\sqrt{1 + 4 + \alpha^2}}\right)$

$$0 = 2 + 8 - 2\alpha$$

$$\alpha = 5$$

28. [4] $\vec{P} \times \vec{Q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & \sqrt{3} & 2 \\ 4 & \sqrt{3} & 2.5 \end{vmatrix} = \sqrt{3} \frac{\hat{i}}{2} + \frac{\hat{j}}{2} - \sqrt{3} \hat{k}$

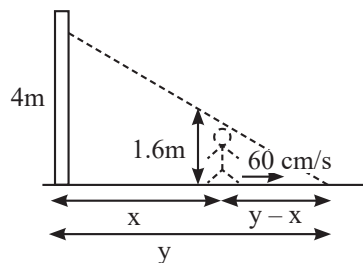
$$\text{and } \Rightarrow |\vec{P} \times \vec{Q}| = \frac{1}{2} \Rightarrow \frac{\vec{P} \times \vec{Q}}{|\vec{P} \times \vec{Q}|} = \frac{1}{2} \left(\sqrt{3} \frac{\hat{i}}{2} + \frac{\hat{j}}{2} - \sqrt{3} \hat{k} \right)$$

$$= \frac{1}{4} (\sqrt{3} \hat{i} + \hat{j} - 2\sqrt{3} \hat{k}) \Rightarrow x = 4$$

29. [180] If angle between them is 180° then

$$\vec{P} \times \vec{Q} = \vec{q} \times \vec{P} = 0$$

30. [40]



From the property of similar triangle,

$$\frac{4}{y} = \frac{1.6}{y-x}$$

$$x = 0.6y$$

$$\frac{dx}{dt} = 0.6 \times \frac{dy}{dt}$$

$$60 = 0.6 \times \frac{dy}{dt}$$

$$\therefore \frac{dy}{dt} = 100 \text{ cm/s}$$

Speed of tip of person's shadow w.r.t person = $100 - 60 = 40 \text{ cm/s}$